Andrew Hicks

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Education

Louisiana State University, Baton Rouge, LA — Aug 2018–May 2024
- Ph.D. in Mathematics
- Concentration: *Computational Mathematics and Numerical Analysis*
- Advisor: Shawn Walker (website)
- GPA: 3.98

Louisiana State University, Baton Rouge, LA — Aug 2018–December 2020
- M.S. in Mathematics
- GPA: 3.96

Ave Maria University, Ave Maria, FL — Aug 2013–May 2017
- B.A. in Mathematics, Economics, *summa cum laude*
- GPA: 3.99

Employment

Carnegie Mellon University, Pittsburgh, PA — August 2024–present
- Postdoctoral Teaching Fellow

Sandia National Laboratories, Albuquerque, NM — Summer 2022, 2023
- NOMAD Research Institute Intern
- Researched interlocking metasurfaces (2023)
- Researched pressure vessel penetration (2022)

Publications

- A. Hicks and Shawn Walker. **"Modeling and Simulation of the Cholesteric Landau-de Gennes Model."** (2024). *Proceedings of the Royal Society A*. 480: 20230813. (link)

Research

"Solving Porous Phase Flow Using Uzawa's Algorithm" Mentor: Noel Walkington — Fall 2025–present
- Summary: Study of porous mediums using implicit Euler, implemented with Uzawa's algorithm

"Numerical Methods for Liquid Crystals and their Optimal Design" Advisor: Shawn Walker — Spring 2020–May 2024
- Summary: Study of the Landau-de Gennes continuum mechanics model for liquid crystals
- Numerical methods: Finite element method, gradient descent, Newton's method

Teaching

Carnegie Mellon University, Pittsburgh, PA

21-256, Multivariate Analysis	Spring 2026
21-671, Computational Linear Algebra	Fall 2025
21-259, Calculus in Three Dimensions	Spring 2025
21-671, Computational Linear Algebra	Fall 2024

Louisiana State University, Baton Rouge, LA

Math 1553, Calculus II Honors	Spring 2024
Math 1550, Calculus I	Fall 2023
Math 1550, Calculus I	Fall 2022
Math 1021, College Algebra	Fall 2019
Math 1431, Business Calculus (recitation)	Spring 2019
Math 1431, Business Calculus (recitation)	Fall 2018

Research mentoring

SUAMI 2025 at Carnegie Mellon University — Summer 2025
Mentored three undergraduate students as part of an REU (link).
Research topic: *Modeling liquid crystals with numerical PDE's*

Directed Reading Program at Louisiana State University — Fall 2023
Mentored one undergraduate student as part of a directed reading program.
Research topic: *The Q-tensor model of Liquid Crystals*

Invited Talks

- **"Modeling and Numerical Analysis of the Cholesteric Landau-de Gennes Model."** 50 minute talk, DePaul University, Chicago, IL (May 16, 2025).
- **"Euclid's *Elements* and the Quadrivium: A Friendly Introduction."** 50 minute talk, University of St. Thomas, Houston, TX (Mar 6, 2024).
- **"Modeling and Numerical Analysis of the Cholesteric Landau-de Gennes Model."** 50 minute talk, Florida Polytechnic University, Lakeland, FL (Feb 28, 2024).

Contributed Talks/Posters

- **"Modeling and Simulation of the Cholesteric Landau-de Gennes Model."** 20 minute talk.
  - SIAM NY-NJ-PA 2025, State College, PA (Nov 1, 2025)
  - SEARCDE 2024, Morgantown, WV (Nov 9, 2025)
  - SIAM TX-LA 2023, Lafayette, LA (Nov 4, 2023)
- **"Dynamic Tailoring of Interlocking Metasurfaces."** Talk, NOMAD 2023 (for Sandia National Laboratories), Albuquerque, NM (Aug 1, 2023).
- **"Modeling and analysis of cholesteric shells."** 20 minute talk.
  - FE Rodeo 2023, College Station, TX (Mar 25, 2023)
  - SCALA 2023, New Orleans, LA (Mar 10, 2023)

- **“Modeling and analysis of cholesteric shells.”** Poster.
  - SIAM TX-LA 2022, Houston, TX (Nov 5, 2022)
  - SIAM Annual Meeting 2022, Pittsburgh, PA (Jul 12, 2022)
- **“Pressure Vessel Enclosure Penetration Energy Prediction.”** Talk, NOMAD 2022 (for Sandia National Laboratories), Albuquerque, NM (Aug 2, 2022).
- **“Python for Beginners.”** Four part, 8 hour lecture series on the Python programming language. Baton Rouge, LA, via Zoom (Oct 18–Nov 8, 2021) (here, under “recent events”)
- **“The History and Ideas Behind Monsky’s Theorem.”** Talk, ASPiRE 2017, Fort Myers, FL (Feb 11, 2017)

Conferences attended

- *SIAM NY-NJ-PA Sectional Meeting 2025*, Penn State University, State College, PA (Oct 31-Nov 2, 2025)
- *SIAM TX-LA Sectional Meeting 2025*, University of Texas at Austin, Austin, TX (Sep 26-28, 2025)
- *Southeastern-Atlantic Regional Conference on Differential Equations* (SEARCDE 2024), West Virginia University, Morgantown, WV (Nov 9-10, 2024)
- *Scientific Computing Around Louisiana* (SCALA 2024), Tulane University, New Orleans, LA (Jan 19-20, 2024)
- *Joint Mathematics Meetings* (JMM24), San Francisco, CA (Jan 2-6, 2024)
- *SIAM TX-LA Sectional Meeting 2023*, University of Louisiana at Lafayette, Lafayette, LA (Nov 3-4, 2023)
- *Finite Element Rodeo* (FE Rodeo 2023), Texas A&M University, College Station, TX (Mar 24-25, 2023)
- *Scientific Computing Around Louisiana* (SCALA 2023), Tulane University, New Orleans, LA (Mar 10-11, 2023)
- *SIAM TX-LA Sectional Meeting 2022*, University of Houston, Houston, TX (Nov 4-6, 2022)
- *SIAM Annual Meeting* (AN22), Pittsburgh, PA (Jul 10-13, 2022)
- *Finite Element Rodeo* (FE Rodeo 2022), Southern Methodist University, Dallas, TX (Mar 4-5, 2022)
- *Scientific Computing Around Louisiana* (SCALA 2020), Louisiana State University, Baton Rouge, LA (Feb 7-8, 2020)
- ICERM workshop: *Numerical Methods and New Perspectives for Extending Liquid Crystaline Systems*, Brown University, Providence, RI (Dec 9-13, 2019)
- *Scientific Computing Around Louisiana* (SCALA 2019), Tulane University, New Orleans, LA (Feb 15-16, 2019)
- *Advancing Student Participation in Research Experiences* (ASPiRE 2017), Florida Gulf Coast University, Fort Myers, FL (Feb 11, 2017)

Programming/software experience

Programming

- Python (highly proficient, software written, 6 hour lecture given)
- C++
- MATLAB
- Linux shell scripting

- MPI
- High Performance Computing (HPC)
- NumPy, SymPy, MatPlotLib
- Git/GitHub/GitLab
- HTML/CSS
- LaTeX

Software
- Firedrake finite element (FE) package
- Abaqus for FE analysis
- LS-Dyna for FE analysis
- AutoCAD
- Microsoft Excel, Word, etc.

Software packages

- **Q-Tensor-3D** (GitHub) – Solves the Landau-de Gennes free energy problem using finite element package Firedrake (Python)
- **SymPyPlus** (GitHub) – Does calculus of variations using SymPy as a base (Python)

Websites designed

- www.dhicksconsulting.com
- www.grecorycc.com

Relevant coursework

Finite Element Methods, Numerical Linear Algebra, Partial Differential Equations, Nonlinear Optimization, Convex Optimization, Machine Learning, Ordinary Differential Equations, Intro to Applied Math, Differential Geometry, Real Analysis, Complex Analysis

Leadership roles

Society for Industrial and Applied Mathematics, LSU Student Chapter (website)

President	Jan 2022–Dec 2023
Webmaster	Jan 2021–Dec 2021
Treasurer	Jan 2020–Dec 2020

Awards/Certifications

- **Dale Carnegie Certificate**: *Effective Communications and Human Relations* (2018)
- **Mathematics Department Award**, Ave Maria University (May 2017)
- **Economics Department Award**, Ave Maria University (May 2017)

Academic interests

Computational mathematics, numerical PDEs, liquid crystals, applied analysis

Foreign languages

- Latin (advanced reading/writing, some conversational proficiency)
- Spanish (beginner/intermediate proficiency reading, writing, speaking)
- Mandarin Chinese (beginner/intermediate proficiency in reading/writing, some conversational proficiency)

Non-math publications

"The Descent of Orpheus." Translation of portion of Ovid's *Metamorphoses* into English. *Contraries* Journal, Fall 2014.